

CXR gRPC Lab Manual

SW.16 — ClaimAnalysis internal RPC + grpcui

CXR Ops Lab — `cxr-ops-lab`

2026-05-31

Abstract

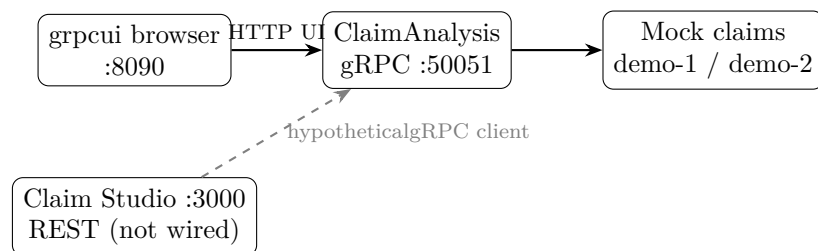
SW.16: `cxr.v1.ClaimAnalysis` gRPC service on **:50051** with `GetClaimStatus` and `AnalyzeClaim`.
Browser UI via **grpcui :8090**. Standalone lab — **Claim Studio (:3000)** keeps REST +
Python spawn.

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Chapter 1

Architecture



Chapter 2

CXR placement (syllabus M2.8)

Today (bootcamp)	Hypothetical gRPC boundary
UI → REST → spawn Python	BFF → AnalyzeClaim → analysis kernel
GraphQL :4000 = external query API	gRPC = internal service-to-service
Kafka :9092 = async events	Publish <i>after</i> analyze; does not replace sync RPC

Chapter 3

Procedure

```
cd cxr-ops-lab
./scripts/19-grpc-up.sh
./scripts/19-grpc-smoke.sh
```

Browser: <http://localhost:8090> → `cxr.v1.ClaimAnalysis` → **AnalyzeClaim** with `lab/grpc/request-`

Chapter 4

Proto (golden RPCs)

```
service ClaimAnalysis {  
  rpc GetClaimStatus(GetClaimStatusRequest) returns (  
    ClaimStatusResponse);  
  rpc AnalyzeClaim(AnalyzeClaimRequest) returns (AnalyzeClaimResponse);  
}
```

Golden: `claim_id = "demo-1" → status = "ok"` (matches SW.15 GraphQL / SW.13 Kafka fixture).

Chapter 5

Files

File	Purpose
lab/grpc/cxr_claim.proto	Service definition
lab/grpc/server.mjs	Node @grpc/grpc-js server
lab/grpc/client-smoke.mjs	Healthcheck + smoke client
compose.grpc.yaml	Server + grpcui
scripts/19-grpc-up.sh	Start stack
scripts/19-grpc-smoke.sh	Verify :8090 + RPC
evidence/SW16-grpc-verify-2026-05-31.md	2026-05-31

Chapter 6

Live adjustments

- **gRPC reflection** — `grpcui` requires reflection; `@grpc/reflection` added to `server.mjs` (without it `grpc-ui` container crash-loops).
- **Healthcheck** — in-container `client-smoke.mjs` (gRPC has no browser GET on `:50051`).
- **grpc-ui restart** — `19-grpc-up.sh` restarts `grpc-ui` after server healthy (avoids stale crash-loop from pre-reflection boot).

Chapter 7

UI evidence — grpcui (:8090)

AnalyzeClaim form for `cxr.v1.ClaimAnalysis`. Golden invoke: `claim_id = demo-1, content = SW.16` golden-path fixture → Response status: `ok`.

Service name: `cxr.v1.ClaimAnalysis` »

Method name: `AnalyzeClaim` ◀

Request Form | Raw Request | Response | History

Request Metadata

Name	Value

+ Add item

Request Data

`cxr.v1.AnalyzeClaimRequest`

claim_id string ☐

content string ☐

Request Timeout

seconds

Invoke

Figure 7.1: grpcui Request Form — **AnalyzeClaim** on **:8090** (user verify 2026-05-31).

Chapter 8

Stop

```
docker compose -f compose.grpc.yaml down
```